# Bonds and yield curves

How do dated payments, expectations and risk premia produce a yield?

# Today's route

Discount single payments, then coupon payments.

Distinguish spot rates, forwards and expectations.

Connect yield changes to bond prices and a curve trade.

LECTURE 3

# Bonds and the yield curve

How do future payments acquire a price today?

# A bond is a schedule of promised payments

$F$: face value, the principal repaid at maturity.

A coupon is an interest payment before or at maturity.

$P$: market price, the amount paid to buy the bond today.

# A coupon rate applies to face value

$$\text{Annual coupon payment} = c_{\text{coupon}} F.$$

$c_{\text{coupon}}$: annual coupon rate in this annual-payment example.

$$c_{\text{coupon}} = 4\%, \quad F = 1{,}000 \quad \Rightarrow \quad c_{\text{coupon}} F = 40.$$

# The three-year cash-flow timeline




At maturity, the investor receives face value plus the final coupon.

# Face value and the price quote are different

Teaching examples often use face value 100 or 1,000.

$$\text{Quote of 98 per 100 face} \quad \Rightarrow \quad \text{980 for 1,000 face.}$$

This example excludes accrued interest. Actual denominations depend on the instrument.

# A zero-coupon bond has one future payment

$R^{\text{spot}}(T)$: annual spot yield on a payment in T years.

$$P = \frac{F}{(1 + R^{\text{spot}}(T))^T}.$$

A spot rate applies from today to the stated maturity.
Annual compounding and certain repayment for this example.

# Discounting a face value of 1,000

$$F = 1{,}000, \qquad T = 2, \qquad R^{\text{spot}}(2) = 5\%.$$

$$P = \frac{1{,}000}{1.05^2} = 907.03.$$

Pay 907.03 now to receive 1,000 in two years.

# A coupon bond discounts every dated payment

$$P = \sum_{k=1}^{n} \frac{c_{\text{coupon}} F}{(1 + R^{\text{bond}})^k} + \frac{F}{(1 + R^{\text{bond}})^n}.$$

| Symbol | Meaning |
|---|---|
| $n$ | Number of annual coupon periods |
| $k$ | Payment-year index |
| $R^{\text{bond}}$ | Single yield to maturity matching the bond price |

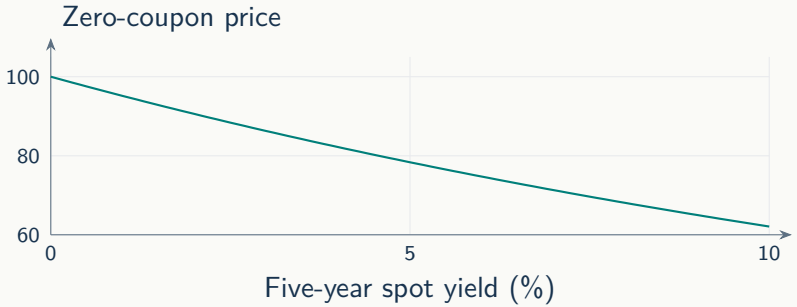
# The coupon-bond calculation

$$P = \frac{40}{1.05} + \frac{40}{1.05^2} + \frac{1{,}040}{1.05^3} = 972.77.$$

A 4% coupon with a 5% yield gives a price below face value.

Annual payments, valued on a coupon date.

# Higher discount yields mean lower bond prices




Hypothetical zero-coupon bond: face value 100, maturity five years, fixed cash flow.

# Duration measures local yield sensitivity

$$\frac{\Delta P}{P} \approx -D_{\text{mod}}\,\Delta R^{\text{bond}}.$$

$D_{\text{mod}}$: modified duration, the bond's local price sensitivity.

Duration 8 and a 0.50-point yield rise
imply approximately $-8 \times 0.005 = -4\%$.

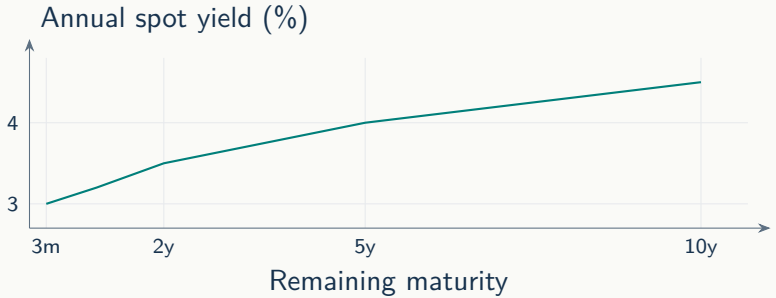
# Treasury bills pay face value without periodic coupons

T-bills are short-term US Treasury securities, with original maturities of one year or less.

$$\text{Pay 98 now} \quad \longrightarrow \quad \text{receive 100 at maturity.}$$

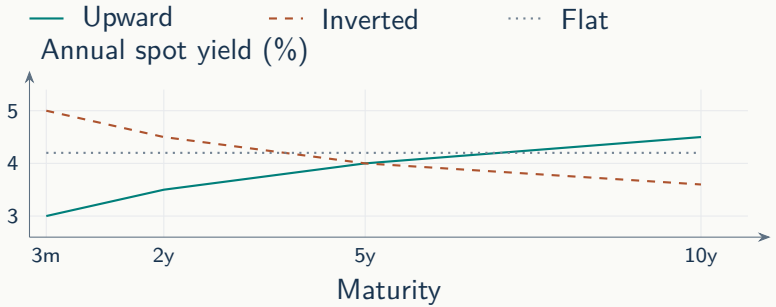
$$\text{Holding-period return} = \frac{100 - 98}{98} = 2.04\%.$$

# The yield curve is one date with many maturities



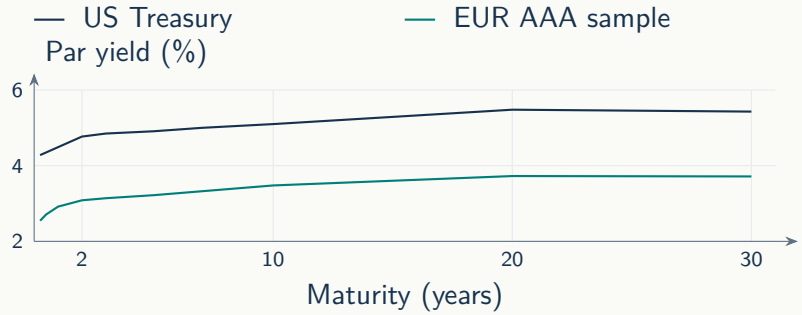

The horizontal axis is maturity. It is not the calendar path of policy rates.

# Yield curves can have different shapes




All shapes are hypothetical. A slope alone is not yet a forecast of future short rates.

# US and euro-area curve snapshot




US: 15 October 2026. EUR: latest verified 14 October. Different currencies and issuer samples.

# Spot, forward and coupon-bond yields

| Rate | What is fixed today? |
|---|---|
| $R^{\text{spot}}(T)$ | Price of a zero-coupon payment at T |
| $R^{\text{forward}}(T_1, T_2)$ | Rate for a future lending interval |
| $R^{\text{bond}}$ | One yield matching a coupon bond's price |

T, T1 and T2 are future dates measured in years from today.

# Two ways to lock a two-year payoff

Route A: buy a two-year zero-coupon bond.

Route B: invest for one year, then reinvest at a second-year rate locked today.

Same initial investment, currency, credit and final payment date.

# No-arbitrage makes the locked payoffs match

$$\boxed{(1 + R^{\text{spot}}(2))^2 = (1 + R^{\text{spot}}(1))(1 + R^{\text{forward}}(1, 2)).}$$

If locked returns differed, borrow along the cheaper route and lend along the more profitable route.

# Calculating the implied forward

$$R^{\text{spot}}(1) = 4\%, \qquad R^{\text{spot}}(2) = 5\%.$$

$$R^{\text{forward}}(1,2) = \frac{1.05^2}{1.04} - 1 = 6.01\%.$$

Both locked routes turn 100 today into 110.25 in two years.

# A locked forward is not a guaranteed forecast

The forward rate is a contract price today.

Next year's realised short rate remains uncertain.

Risk premia can separate an implied forward from the physical expected future rate.

# A long yield contains a path and a premium

Expected average short rate over the bond's life.

Plus a term premium for holding long-duration exposure.

$$3.0\% + 0.5\% = 3.5\% \quad \text{in a hypothetical example.}$$

# Writing the yield decomposition mathematically

$$R^{\text{spot}}(n) \approx \frac{1}{n} \sum_{k=0}^{n-1} \mathbb{E}_t R_{t+k}^{\text{policy}} + TP(n).$$

| Symbol | Meaning |
|---|---|
| $n$ | Maturity in years |
| $\mathbb{E}_t$ | Expectation formed today |
| $TP(n)$ | Term premium for that maturity |

# Today's rate and a ten-year yield can move differently

A cut today changes only part of the expected future path.

Later expected rates or the term premium can rise enough to raise the long yield.

A policy decision does not fix the entire yield curve.

# Exercise: a cut today with a bond sell-off

Expected average short rates over ten years rise 0.10 points.

The ten-year term premium rises 0.20 points.

Predict the yield move and the duration-8 price change.

# The decomposition gives the answer

$$\Delta R^{\text{spot}}(10) \approx 0.10 + 0.20 = 0.30 \text{ percentage points.}$$

$$\frac{\Delta P}{P} \approx -8 \times 0.003 = -2.4\%.$$

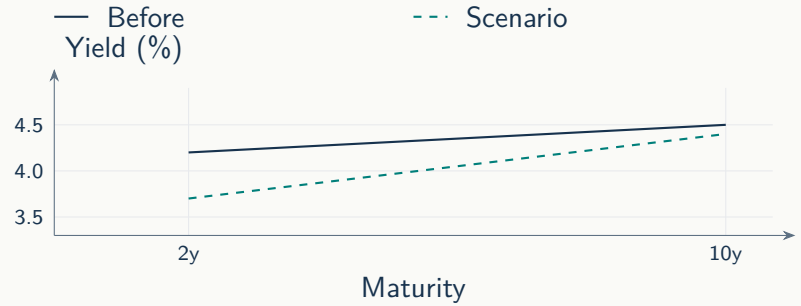
The long yield rises and the bond loses value in this specified scenario.

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# A curve-slope exercise

How can two bond positions express a relative rate view?

# A curve view needs two maturity forecasts




Both yields fall. The two-year yield falls more. Hypothetical matched bond-yield moves.

# This is a bull steepening

$$\text{Slope} = \text{ten-year yield} - \text{two-year yield}.$$

$$0.30 \text{ points} \quad \longrightarrow \quad 0.70 \text{ points}.$$

Bull: yields fall and bond prices rise.
Steepening: the long-minus-short yield spread increases.

# Expressing the relative view with bonds

Buy the two-year bond: benefit if its yield falls.

Short the ten-year bond: lose if its yield falls.

Size the legs so the stronger two-year rally
can outweigh the ten-year loss.

# DV01 puts the two legs on comparable terms

One basis point (bp) is 0.01 percentage point, or 0.0001.

$$DV01 \approx VD_{\text{mod}} \times 0.0001.$$

$V$: current market value.
DV01: approximate currency price change for a one-bp yield move.

# Exercise: choose the size of the short leg

| Position | Market value | Duration | DV01 |
|---|---|---|---|
| Long two-year | 1,000,000 | 1.8 | 180 / bp |
| Short ten-year | ? | 8.0 | 180 / bp |

What ten-year market value matches the long leg's DV01?

# Equal DV01 requires unequal market values

$$V_{\text{short}} = \frac{180}{8 \times 0.0001} = 225{,}000.$$

| Position | Market value | Signed DV01 |
|---|---|---|
| Long two-year | 1,000,000 | +180 / bp |
| Short ten-year | 225,000 | -180 / bp |

Approximately neutral to a small parallel yield move at inception.

# Work out the scenario profit and loss

| Leg | Yield move | Approximate P&L |
|---|---|---|
| Long two-year | -50 bp | $+180 \times 50 = +9{,}000$ |
| Short ten-year | -10 bp | $-180 \times 10 = -1{,}800$ |
| Combined | 40 bp steepening | $+7{,}200$ |

Local duration calculation, before carry, financing, costs and convexity.

# Counterfactual: what would make this trade lose?

The ten-year minus two-year yield spread narrows instead.

An expected cut is already priced, so the curve does not move as hoped.

Financing, carry and imperfect hedging can overwhelm the gain.

# Further reading

Dmitriy Sergeyev, Bocconi 30054 course materials.

Krugman, Obstfeld & Melitz, *International Economics*.

Source links and worked answers accompany each slide in the notes.